Hamidreza Raei

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Research Interests

- **Reinforcement Learning for Manipulation**
- **Data-Driven/model-based Control**

Education

Polytechnic University of Milan — Milan
PhD Department of Electronic, Information and Bio-engineering — *Sep 2023 – Expected Dec 2026*
GPA so far (29.66/30)

Gwangju Institute of Science and Technology — Gwangju
MSc Mechanical Engineering; ***Thesis: Camera and LiDAR fusion for autonomous landing on moving target*** — *Aug 2022*
GPA (3.5/4.0)

Isfahan University of Technology — Isfahan
BSc Mechanical Engineering ; ***Thesis: Controller Design for a Ball and Beam system using time-response analysis*** — *June 2020*
last two year GPA (3.46/4.0)

Publications

- **H.Raei, J.Zhao, E.De Momi, A.Ajoudani: "MonoFold: Monocular Sim-to-Real Manipulation of Deformable Sheets with Compact Geometric Representations"**
  Published for IEEE Robotics and Automation Letters (RA-L) — Oct 2026
  DOI: 10.1109/LRA.2026.3746129

- **Changda Tian∗, Hamidreza Raei∗, Arash Ajoudani, and Panos Trahanias: "Sim-to-Real Reinforcement Learning for Ball-Balancing Locomotion on Quadruped Robots"**
  Published for IEEE International Conference on Advanced Intelligent Mechatronics // * **Co-First Author** — May 2026
  DOI: 10.1109/AIM65483.2026.11658359

- **H.Raei, E.De Momi, A.Ajoudani: "A Reinforcement Learning Approach to Non-prehensile Manipulation through Sliding"**
  Published for IEEE Robotics and Automation Letters (RA-L) — Jun 2025
  DOI: 10.1109/LRA.2025.3578238

- **H.Raei, J.Gandarias, E.De Momi, P. Balatti, A. Ajoudani: "A Multipurpose Interface for Close- and Far-Proximity Control of Mobile Collaborative Robots"**
  Published BioRob Conference 2024 proceeding — Sep 2024
  DOI: 10.1109/BioRob60516.2024.10719833

- **H.Raei, Y.Cho, K.Park: "Autonomous landing on moving targets using LiDAR, Camera and IMU sensor Fusion"**
  Published in proceedings of the 13th Asian Control Conference (ASCC 2022, International Conference) — Mar 2022
  DOI: 10.23919/ASCC56756.2022.9828342

Honor and Awards

- *Best Presentation Award,* ***TAROS conference 2025*** *at York University, United Kingdom* — *Aug 2025*
- *Awarded the prestigious* ***Marie Skłodowska-Curie Actions*** *scholarship for doctoral studies. https://raicam.eu/* — *Sep 2023*

Skills

- **Programming Languages**: Python, C++, MATLAB
- **Machine Learning**: PyTorch, TensorFlow, NumPy, OpenCV, Weights & Biases; deep reinforcement learning (SAC, DDPG, PPO), CNN/LSTM architectures, domain randomization, sim-to-real transfer
- **Simulation & Robotics**: MuJoCo, Isaac Sim, Isaac Lab, Genesis, ROS1, ROS2, PX4, MATLAB & Simulink, CATIA
- **Development Tools and version Control**: Docker, Git, Linux, Bash, SLURM/HPC cluster training

Bootcamps and Competetions

Zhejiang University Global Innovation Practice Program (Industrial AI and Robotics) — Hangzhou, China
https://www.zju.edu.cn/english/GlobalInnovation/main.htm — *Jul 2026*

ENRICH 2025 Robotics Hackathon (Team RAICAM) — Vienna, Austria
https://enrich.european-robotics.eu/ — *Jun 2025*

Multi-robot mapping, and autonomous navigation and coordinated valve closure at ENSTA Paris — Paris, France
https://raicam.eu/ — *Nov 2024*

Multi robot autonomous navigation, and teleoperated inspection by Cobots — Genova, Italy
https://raicam.eu/ — *March 2024*

Robothon® - The Grand Challenge 2023 (Team Upcyclers) — Munich, Germany
https://automatica-munich.com/en/munich-i/robothon/teams/ — *March 2023*

We were ranked 4th out of 20 teams

Work Experience

- **Research Fellow in Human-Robot Interfaces and Interaction Lab at IIT**: **https://hri.iit.it/** — Dec 2022 - So Far
  - Built an end-to-end **monocular perception-to-action pipeline** for deformable sheet manipulation (SAM2 segmentation, Depth Anything V2, ResNet state regressor) that compresses raw RGB into a **9-D corner-relative state**, achieving **0.038 m** real-world corner-estimation error against **0.031 m** in simulation over 12 marker-tracked trajectories.
  - Designed a **multi-fidelity RL training pipeline** in MuJoCo with a mixture-of-experts policy; validated by **270 simulated and 36 hardware trials per manipulation primitive** across 3 sheet materials and 3 grasp configurations.
  - Developed a **DDPG actor-critic** controller for non-prehensile sliding manipulation with LSTM-based real-time friction estimation and domain randomization, achieving **zero-shot sim-to-real transfer** to a Franka Emika Panda (**IEEE RA-L 2025**).
  - Developed **sim-to-real RL for ball-balancing locomotion** on a quadruped robot, transferring GPU-accelerated simulation training to hardware (**IEEE AIM 2026**).
  - Implemented a **visual-odometry teleoperation interface** for a 7-DoF arm, estimating operator hand pose from RGB-D and IMU data with **no environment-mounted sensors**, and a multipurpose close-/far-proximity control interface for mobile collaborative robots (**BioRob 2024**).
  - Delivered **4 peer-reviewed publications** (2 IEEE RA-L, 2 conference) and represented the lab in 4 international robotics competitions and hackathons.
- **Research Assistant in the Dynamics and Control lab at GIST**: **https://sentor.gist.ac.kr/sentor/** — Sep 2020 - Nov 2022
  - Led the **multi-sensor fusion** effort (LiDAR, camera, IMU) for autonomous UAV landing on a **moving** target, taking the system from simulation to outdoor flight tests and first-author publication (**ASCC 2022**).
  - Owned the full stack: 30 Hz LiDAR/camera/IMU relative localization, Kalman-filter state estimation, and benchmarked PID, fuzzy-PID and LQR position/altitude controllers on PX4 hardware.
  - Teaching assistant for **System Modeling and Control**, supervising MATLAB/Simulink controller implementations for graduate students.
- **Internship at Shahid Mohammad Montazeri Power plant**: **https://www.montazeri-ccpp.ir/en/** — Jan 2018 - Aug 2018
  - Shadowed turbine and boiler maintenance operations in a combined-cycle plant; documented instrumentation and control loops for rotating machinery.

Teaching Experience

- **Teacher Assistant for System Modeling and Control course**: — Sep 2021 - Dec 2021
  Under the supervision of professor Kyihwan Park
  - **Teaching student** implementation of various system and their controllers in matlab (simulink).
- **Teacher Assistant for Entrepreneurship Course** : — Feb 2019 - Aug 2019
  Under the supervision of Dr.Alinaghian and Dr.Reisi-Nafchi
  - **facilitating projects:**Helping the students of this course at the beginning to understand bases of the Entrepreneurship and then Helping them to make strategies for SWOT Matrix

REFERENCES

- **Director of Human-Robot Interfaces and Interaction (HRI²) laboratory at IIT, Arash Ajoudani - Email: Arash.Ajoudani@iit.it**
- **Professor of Electronic Information and Bioengineering Department, Elena De Momi - Email: elena.demomi@polimi.it**
- **Professor of Mechanical Engineering department (GIST), Kyihwan Park - Email: khpark@gist.ac.kr**
  MSc Thesis advisor